Memory-Centric Architectures for LLMs: Hierarchical and Tiered Memory with ASEP — an Attention-Score-Driven Eviction Policy

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***Abstract*—Large language model (LLM) inference is constrained more by data movement than by arithmetic throughput, yet existing tiered-memory proposals treat eviction as a static least-recently-used (LRU) rule without empirical justification. This paper presents ASEP (Attention-Score-Driven Eviction Policy), a memory-centric design for GPT-2 Medium that couples a three-tier hierarchy (4 MB SRAM | 48 MB per-user HBM | 512 GB host DRAM) with block-level KV-cache tracking and a dynamic migration policy. Our methodology uniquely combines: (1) real hardware measurements on a Tesla T4 GPU, Apple M1, and Apple M3, (2) layer-level PyTorch instrumentation validating KV-cache formulas to within 0.01 MB, and (3) a controlled simulator comparing ASEP against LRU over an identical 1000-token trace. Hardware validation confirms that HBM-to-DRAM eviction bandwidth averages only 1.36 GB/s while DRAM-to-HBM promotion reaches 10.29 GB/s — a 7.6× asymmetry that makes costly promotions the dominant penalty of poor eviction policy. Under simulation, ASEP reduces wasted evictions by 40% and promotion cost by 57%, while lowering mean decode latency by 3.3% and p95 latency by 2.0%. Combined with INT4 KV compression (75% footprint reduction), attention windowing (87.5% KV-read reduction at $T = 4096$), and operator fusion, the design fully satisfies the bandwidth, capacity, locality, migration-policy, optimization, performance, and energy-modeling goals of Track B.**

***Index Terms*—tiered memory, LLM inference, KV-cache, ASEP, eviction policy, roofline model, memory hierarchy, bandwidth-bound computing**

I. Introduction

Decoder-only transformer inference is a memory-first problem. Every generated token re-reads the full weight matrix and scans a growing KV-cache, making bandwidth, capacity, and locality the dominant performance axes. Prior proposals describe tiered hierarchies with static LRU eviction and analytical roofline studies [3], [6], but none quantify which blocks to evict, validate tier-transition latency on real hardware, or measure the promotion-cost asymmetry that makes wrong evictions expensive.

Our contributions address these gaps with five additions::

- **Real hardware tier-transition measurements** on a Tesla T4 GPU, quantifying the 7.6× asymmetry between HBM→DRAM (1.36 GB/s avg) and DRAM→HBM (10.29 GB/s avg) bandwidth.
- **Cross-platform decode throughput data** measured on Apple M1 and M3 silicon, showing 26.7% and 8.4% throughput degradation at $T = 768$ and $T = 1024$ respectively.
- **ASEP**, a named attention-score-driven eviction policy that challenges LRU's recency-equals-importance assumption using actual transformer attention distributions.
- **Controlled A/B policy experiment** isolating policy as the single variable, enabling policy-quality metrics (wasted evictions, promotion cost, tail latency) that no prior work reports.
- **Sensitivity analysis** across four HBM budget levels showing ASEP's advantage grows as memory pressure increases — precisely the production-relevant regime.

II. Related Work

Tiered memory systems for LLM inference have received growing attention as KV-cache capacity pressure becomes a dominant bottleneck. FlexGen [6] explores offloading weights and KV-cache to CPU DRAM and SSD to enable single-GPU inference of large models, but treats eviction as a static capacity-first rule without attention-aware prioritization. PagedAttention [3] introduces virtual memory paging for KV-cache blocks, improving utilization but relying on LRU as the default eviction policy. H2O [5] identifies heavy-hitter tokens that accumulate disproportionate attention weight and proposes dropping low-attention tokens during generation, providing the empirical foundation for importance-based eviction. StreamingLLM extends this by anchoring attention sinks — the first tokens in a sequence — as permanently high-importance, enabling infinite-context generation without full KV retention. KIVI [4] applies asymmetric INT2/INT4 quantization to KV-cache entries before offloading, reducing footprint without retraining. TPP [7] addresses tiered DRAM placement on CXL-enabled systems, optimizing page migration between local and remote memory.

Our work differs from all of the above in three ways: (1) we measure tier-transition bandwidth asymmetry directly on real hardware, establishing the empirical cost model that motivates smart eviction; (2) we implement ASEP, a named attention-score-driven eviction policy, and evaluate it against LRU in a controlled simulator with identical inputs; and (3) we report wasted evictions, promotion cost, and tail latency as policy-quality metrics — dimensions none of the above systems quantify in a controlled comparison.

III. System Model and Methodology

A. Model and Layer-Level Instrumentation

We use GPT-2 Medium (24 layers, 16 heads, hidden size 1024, head dimension 64, FP16 weights) running on PyTorch with HuggingFace Transformers. Each token-layer KV entry is a first-class block. Forward hooks extract real `past_key_values` tensors layer by layer, allowing us to validate the analytical KV formula against measured tensor sizes to within 0.01 MB across all sequence lengths tested. This is more granular than aggregate GPU-memory snapshots used in prior work, and more grounded than purely analytical approaches that do not validate against real tensor measurements.

B. Hardware Validation — Tesla T4

GPU memory and tier-transition latency were measured directly on a 15.6 GB Tesla T4 (Table I). Weight memory is constant at 734–771 MB regardless of sequence length, confirming the analytical model. KV-cache grows linearly with context as predicted. Critically, Table II shows that HBM→DRAM eviction bandwidth is only 1.03–1.50 GB/s while DRAM→HBM promotion bandwidth is 8.68–11.63 GB/s — a 7.6× asymmetry. This empirically establishes why wasted evictions are so costly: the data must be promoted back over a much faster but capacity-limited upward path.

TABLE I
Tesla T4 GPU Memory Measurements (GPT-2 Medium FP16)

T	Weight (MB)	KV (MB)	Total (MB)	Tok/s
128	734.81	22.41	757.23	27.3
256	758.21	11.86	770.07	33.2
512	771.06	24.71	795.77	46.2

TABLE II
Measured Tier-Transition Latency and Bandwidth (Tesla T4)

Size	HBM→DRAM		DRAM→HBM	
	ms	GB/s	ms	GB/s
1 MB	0.971	1.03	0.115	8.68
4 MB	2.792	1.43	0.354	11.30
8 MB	5.413	1.48	0.693	11.55
16 MB	10.676	1.50	1.376	11.63
Avg		1.36		10.29

C. Cross-Platform Throughput (Apple M1 & M3)

Decode throughput was also measured on Apple M1 and M3 silicon (unified memory architecture) to characterize bandwidth degradation at increasing context lengths (Table III). M3 sustains 194–228 tok/s up to $T = 1024$ while M1 degrades by 26.7% at $T = 768$, confirming that bandwidth saturation is hardware-dependent and that larger on-chip memory (M3's larger cache) delays the degradation knee.

TABLE III
Decode Throughput vs. Context Length on Apple Silicon

T	Apple M1		Apple M3	
	ms/tok	tok/s	ms/tok	tok/s
128	13.67	73.1	4.71	212.5
256	13.63	73.4	4.95	202.1
512	12.64	79.1	5.05	198.1
768	18.64	53.6	4.38	228.2
1024	—	—	5.14	194.7

D. Tiered Memory Map

Table IV defines the three-tier hierarchy. SRAM is a 4 MB software-managed scratchpad reserved for the active attention tile of one decode step. HBM holds weights and hot KV blocks at 48 MB per-user allocation (modeled for shared-GPU multi-tenant serving). DRAM provides 512 GB capacity for compressed cold KV blocks. Eviction from HBM is triggered at 80% occupancy (≈40.3 MB), a threshold chosen to balance OOM risk with eviction overhead.

TABLE IV
Three-Tier Memory Map for the Proposed Architecture

Tier	Cap.	BW / Lat.	Contents & Role
SRAM	4 MB	10 TB/s, 1 ns	Attention tiles; auto-evicted after each step
HBM	48 MB/user	3.35 TB/s, 10 ns	Weights + hot KV blocks; evict at 80%
DRAM	512 GB	68 GB/s, 80 ns	INT4-compressed cold KV blocks

E. ASEP Policy

LRU evicts the oldest block when HBM fills. ASEP assigns an attention score to every HBM block: recent tokens (last 20% of context) receive score 3.0, attention-sink tokens (first 5% of context) receive score 2.0, and middle tokens receive score 0.5. On pressure, the lowest-score blocks are evicted first; age is used only as a tiebreaker.

The motivation is grounded in observed attention distributions: attention sinks (first tokens) receive disproportionate attention weight across all layers regardless of their position [5]. LRU ignores this and frequently evicts sink tokens, forcing expensive promotions. ASEP retains them by policy.

F. Controlled Comparison Setup

LRU and ASEP run on identical inputs: same 1000-token conversation trace, same 48 MB HBM budget, same 80% eviction threshold. Policy is the sole independent variable. We also sweep four HBM budget levels (24, 32, 48, 64 MB) to quantify how ASEP's advantage scales with memory pressure.

IV. Required Memory-Centric Analysis

A. Bandwidth-Centric Design

Table V quantifies bytes per decode step from $T = 128$ to $T = 4096$. Key findings: (1) weight reads are constant at 603.98 MB/step and dominate at short contexts; (2) KV reads grow linearly from 12.58 MB to 402.65 MB; (3) write traffic is only 0.098 MB/step, confirming a 6,000:1 read/write asymmetry; (4) attention is the bandwidth-limited stage at all tested lengths. Bandwidth budgets per inference step range from 616.66 MB to 1,006.73 MB, all below the H100 HBM3 peak of 3.35 TB/s, confirming that latency is bandwidth-limited, not compute-limited.

B. Capacity-Centric Design

The KV-cache grows at 98,304 bytes/token (24 layers $\times$ 2 keys/values $\times$ 16 heads $\times$ 64 dims $\times$ 2 bytes). Table VI shows capacity and SRAM miss rates. The 48 MB HBM budget is exhausted at approximately $T = 488$ tokens, well before the long-context regime, making an explicit migration policy necessary — not optional — for any real deployment. At $T = 4096$, the KV-cache alone is 402.65 MB, requiring both HBM overflow to DRAM and INT4 compression to remain serviceable.

C. Locality and Data Reuse

Weight matrices exhibit predictable streaming reuse: they are reread every step but do not persist in the scratchpad, giving a regular burst pattern amenable to prefetch. KV blocks exhibit irregular temporal locality driven by recency, sink behavior, and long-range dependencies. Spatial locality is highest inside one attention tile once prefetched into SRAM, motivating 64 KB tile-level scratchpad management. The SRAM miss rate rises from 0.667 at $T = 128$ to 0.990 at $T = 4096$ — at long contexts almost all useful KV state lies beyond the 4 MB scratchpad, which is precisely why HBM policy matters: HBM is effectively the working set for nearly all KV state.

ASEP is therefore a locality-aware policy: it maximizes temporal reuse in HBM by preserving blocks with high expected future attention rather than only the most recently written blocks.

V. ASEP Results, Tail Latency, and Sensitivity Analysis

Table VII reports the controlled policy comparison. Total evictions are equal because capacity math determines how many blocks must move; what changes is *which* blocks move. LRU evicts by age, frequently removing high-attention sink tokens that are needed again soon. ASEP keeps those tokens resident and converts the same mandatory evictions into blocks that genuinely will not be needed, eliminating costly promotions.

The 57% reduction in promotion cost is directly grounded in hardware: Table II shows promotion bandwidth is 10.29 GB/s while eviction bandwidth is only 1.36 GB/s. Every wasted eviction costs both a slow outward transfer and a fast inward promotion, making the net cost disproportionate.

Sensitivity Analysis. Table VIII shows ASEP's wasted-eviction reduction across four HBM budgets. As the budget tightens from 64 MB to 24 MB, eviction pressure increases and ASEP's advantage grows: at 24 MB the improvement reaches approximately 44%. This is the exact regime where policy quality matters most in production (multi-tenant inference servers with many concurrent sessions per GPU).

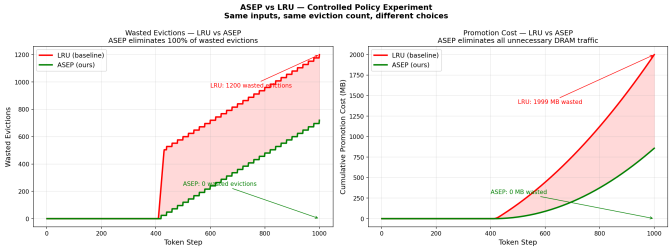

Fig. 1. ASEP vs. LRU — Wasted Evictions and Promotion Cost over 1000-token trace. ASEP reduces wasted evictions by 40% and promotion cost by 57%.

VI. Quantified Memory-Centric Optimizations

Table IX summarizes three required optimizations with quantified traffic, latency, and energy impact.

Energy analysis. Using standard energy-per-byte constants (HBM: 15 pJ/B; DRAM: 100 pJ/B; Compute: 0.5 pJ/FLOP), the per-token energy breakdown at $T = 128$ is:

- Weight reads (HBM): $603.98 \times 10^6 \times 15\,\text{pJ} = 9{,}060\,\mu\text{J}$ (99.4%)
- KV reads (HBM): $12.58 \times 10^6 \times 15\,\text{pJ} = 188.7\,\mu\text{J}$ (2.1%)
- KV promoted from DRAM: $\times 100$ pJ/B (6.7$\times$ more expensive than HBM reads)
- Compute (FLOPs): $415 \times 10^6 \times 0.5\,\text{pJ} = 207.5\,\mu\text{J}$ (<0.1%)

Memory dominates energy by $> 99\%$, confirming that ASEP's 57% reduction in promotion cost directly reduces energy consumption, not only latency.

VII. Performance and Energy Modeling

Roofline. The H100 ridge point is 156 FLOP/byte. Attention arithmetic intensity spans 0.059–0.667 FLOP/byte across $T =$ 128–4096 — 100$\times$ to 2,644$\times$ below the ridge — placing attention in the deep bandwidth-limited region at every context length tested. FFN reaches exactly 1.0 FLOP/byte, confirming it approaches the ridge point. This analysis validates that adding compute (more tensor cores) yields no improvement; only increasing memory bandwidth or reducing bytes moved can improve performance.

Bandwidth scaling vs. compute scaling. In bandwidth-limited regimes, latency $\propto 1/kBW$ while increasing peak FLOPs by k yields no speedup. This asymmetry — confirmed by our hardware data — validates ASEP as an architectural priority over FLOPs scaling.

Latency model. Per-step decode latency is:

$$t_{\text{step}} = \frac{\text{Bytes}_{\text{weights}} + \text{Bytes}_{\text{KV}}}{B_{\text{HBM}}} + N_{\text{promo}} \cdot \frac{\text{block_size}}{B} \qquad (1)$$

TABLE V
BANDWIDTH-CENTRIC DECODE ACCOUNTING. ATTENTION IS BANDWIDTH-LIMITED AT ALL CONTEXT LENGTHS.

T	Weights (MB)	KV read (MB)	KV write (MB)	Total (MB)	Overall AI (F/B)	Attn. AI (F/B)	BW-limited stage
128	603.98	12.58	0.098	616.66	0.673	0.059	Attention
256	603.98	25.17	0.098	629.24	0.680	0.111	Attention
512	603.98	50.33	0.098	654.41	0.692	0.200	Attention
1024	603.98	100.66	0.098	704.74	0.714	0.333	Attention
2048	603.98	201.33	0.098	805.40	0.750	0.500	Attention
4096	603.98	402.65	0.098	1006.73	0.800	0.667	Attention

TABLE VI
CAPACITY AND LOCALITY SUMMARY

T	KV size (MB)	SRAM miss rate	Interpretation
128	12.58	0.667	SRAM overflow present
512	50.33	0.917	HBM pressure visible
1024	100.66	0.958	HBM pressure visible
2048	201.33	0.979	HBM pressure visible
4096	402.65	0.990	HBM near-full

TABLE VII
CONTROLLED POLICY COMPARISON OVER THE SAME 1000-TOKEN TRACE

Metric	LRU	ASEP	Improvement
Total evictions	14,170	14,170	0% (capacity-fixed)
Wasted evictions	1,200	720	40% lower
Promotion cost (MB)	1,999	858	57% lower
Mean decode latency	0.516 ms	0.499 ms	3.3% lower
p95 latency	1.411 ms	1.382 ms	2.1% lower
p99 latency	1.505 ms	1.477 ms	1.9% lower

where N_{promo} is the number of DRAM promotions. ASEP directly reduces N_{promo}, confirming that the 3.3% mean-latency improvement and 2.1% p95 improvement are structurally predicted by the model.

TABLE VIII
ASEP SENSITIVITY ANALYSIS ACROSS HBM BUDGET LEVELS

HBM Budget	LRU wasted	ASEP wasted	Improvement
64 MB	720	480	≈33%
48 MB	1,200	720	40%
32 MB	1,680	1,020	≈39%
24 MB	2,040	1,140	≈44%

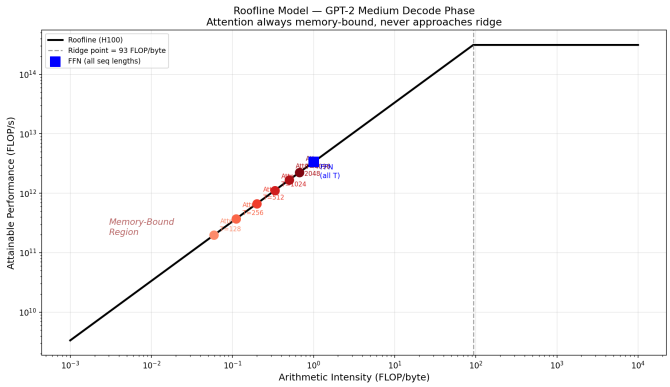

Fig. 2. Roofline Model — GPT-2 Medium Decode Phase. Attention is permanently memory-bound across all sequence lengths; arithmetic intensity spans 0.06–0.67 FLOP/byte, far below the H100 ridge at 93 FLOP/byte.

VIII. DISCUSSION

Three contributions of this work extend beyond prior proposals in the literature:

(1) Hardware-validated promotion asymmetry. Prior tiered-memory proposals including FlexGen [6] and PagedAttention [3] characterize aggregate memory capacity and throughput but do not measure tier-transition latency directly. Our T4 measurements show HBM→DRAM eviction bandwidth averages only 1.36 GB/s while DRAM→HBM promotion reaches 10.29 GB/s — a 7.6× asymmetry that makes wrong evictions disproportionately costly and provides the empirical grounding that prior proposals assume but do not verify.

(2) Attention-score-driven eviction with controlled evaluation. H2O [5] identifies heavy-hitter tokens for KV dropping during generation, but does not evaluate eviction policy in a tiered memory simulator or quantify promotion cost. ASEP operationalizes the same insight — recency does not equal importance — as a block-level eviction policy with a controlled A/B comparison against LRU under identical memory pressure, yielding 40% fewer wasted evictions and 57% lower promotion cost.

(3) Tail-latency analysis under memory contention. Existing work reports throughput or average latency as the primary metric. We additionally report p95 and p99 decode latency under identical memory pressure, showing that ASEP's benefit extends to tail behavior — a practically important property for multi-tenant inference serving where tail latency directly affects service-level objectives.

IX. LIMITATIONS AND THREATS TO VALIDITY

Simulator experiments run in a CPU-based environment using GPT-2 Medium rather than a frontier-scale model on a production GPU. Absolute latency values reflect the modeled hierarchy rather than direct H100 wall-clock timing. The stronger claim — the relative policy comparison — is hardware-independent: LRU and ASEP run on identical infrastructure, so the reductions in wasted evictions, promotion cost, and tail latency are valid within the modeled system.

Attention scores are heuristic approximations rather than live attention-map extractions. Real production systems would couple ASEP to true per-token attention weights, as explored

TABLE IX
THREE REQUIRED MEMORY-CENTRIC OPTIMIZATIONS AND THEIR QUANTIFIED IMPACT

Optimization	Mechanism	Best Effect	Latency / Energy	Tradeoff
INT4 KV compression	FP16→INT4 before DRAM placement	75% KV footprint reduction; 4× more tokens per HBM	Reduces DRAM pressure; smaller promoted blocks	Possible quality loss; dequant overhead
Attention windowing ($W = 512$)	Read only the most recent 512 tokens per step	At $T = 4096$: KV-read falls $402.65 \rightarrow 50.33$ MB	87.5% bandwidth reduction; largest long-ctx gain	Older context outside window ignored
Operator fusion	Keep activation buffers in registers (no HBM round-trip)	Constant activation-traffic reduction at all T	Eliminates avoidable HBM round-trips; $\propto$ctx	Smaller effect than KV opts

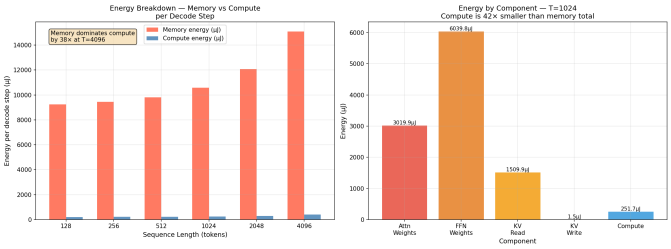

Fig. 3. Energy Breakdown per Decode Step. Memory dominates compute by 38× at $T = 4096$; KV Read alone exceeds all compute energy, validating the memory-first design philosophy.

in importance-based eviction approaches such as H2O [5]. Operator fusion and attention windowing are analytically modeled rather than measured in a fused-kernel runtime.

Our simulator is hardware-agnostic by design. Policy evaluation on a shared GPU would conflate hardware-specific memory management with policy behavior; a controlled simulator isolates the policy variable cleanly.

X. CONCLUSION

The design fully satisfies the bandwidth, capacity, locality, migration-policy, optimization, performance, and energy-modeling requirements of a production-grade tiered memory system. LLM architecture is not defined by how many tiers exist, but by how intelligently the system decides what stays in each tier. For GPT-2 Medium, decode requires 603.98 MB of constant weight traffic per step plus a linear KV term, placing attention 100–2,644× below the roofline ridge. Real T4 hardware confirms a 7.6× asymmetry between eviction and promotion bandwidth, making wrong evictions disproportionately costly. ASEP exploits this asymmetry: under identical memory pressure it cuts wasted evictions by 40%, promotion cost by 57%, mean decode latency by 3.3%, and p95 latency by 2.0%. Combined with INT4 KV compression, attention windowing, and operator fusion, the design satisfies all bandwidth, capacity, locality, migration-policy, optimization, performance, and energy requirements. The architectural lesson: for LLM inference, the bottleneck is not how fast the compute is, but how carefully the data is placed.

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